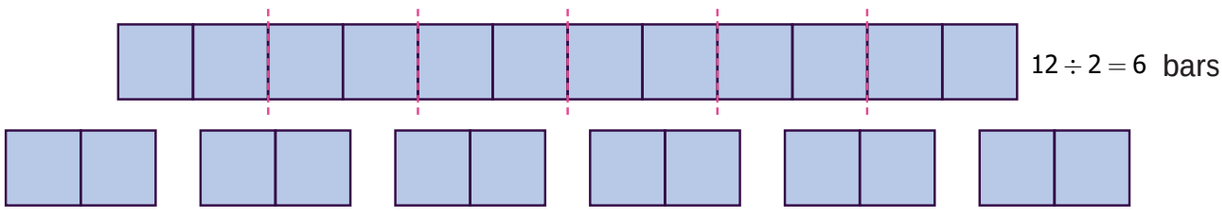
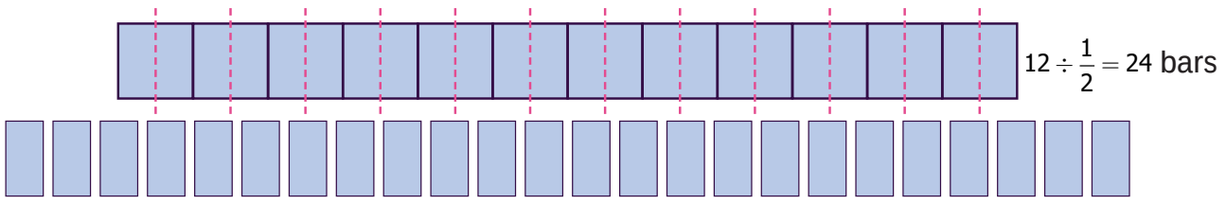
Here, we can observe that, dividing a whole number 6 by a fraction $\frac{1}{2}$ is the same as multiplying a whole number 6 by 2, where 2 is the ***reciprocal*** of $\frac{1}{2}$. Generally, dividing a number by a fraction is the same as multiplying that number by the reciprocal of the fraction.

Let us discuss the same situation in another way. Let us take a bar of length $12\,cm$

How many $2\,cm$ bars are there in a $12\,cm$ bar?



Now find how many $\frac{1}{2}$ cm bars are there in a $12\ cm$ bar?



Let us observe and complete the following:

i) $\frac{3}{7} \times \frac{7}{3} = \frac{21}{21} = 1$ ii) $\frac{1}{9} \times 9 = \frac{9}{9} = 1$ iii) $8 \times \frac{1}{8} = \frac{8}{8} = ?$ iv) $\frac{13}{4} \times ? = 1$ v) $\frac{4}{3} \times ? = 1$

From the above, we can see that ***the Product of a fraction and its reciprocal is always 1***.

Example 19 Kandan shares $\frac{1}{2}$ piece of a cake between 2 persons. What will be the share of each?

Solution To know the share, we need to find $\frac{1}{2} \div 2$

$$\frac{1}{2} \div 2 = \frac{1}{2} \times \frac{1}{2} \quad \left(\text{reciprocal of 2 is } \frac{1}{2}\right)$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1 \times 1}{2 \times 2} = \frac{1}{4}$$

Try these

i) How many 6s are there in 18?

ii) How many $\frac{1}{4}$ s are there in 5?

iii) $\frac{1}{3} \div 5 = ?$

Example 20 Divide $4\frac{1}{2}$ by $3\frac{1}{2}$

Solution

$$4\frac{1}{2} \div 3\frac{1}{2} = \frac{9}{2} \div \frac{7}{2}$$

$$= \frac{9}{2} \times \frac{2}{7} \quad \left(\text{reciprocal of } \frac{7}{2} \text{ is } \frac{2}{7}\right)$$

$$= \frac{9}{7}$$

Example 21 An oil tin contains $7\frac{1}{2}$ litres of oil which is poured in $2\frac{1}{2}$ litres bottles. How many bottles are required to fill $7\frac{1}{2}$ litres of oil?

Solution The number of bottles required $= \frac{15}{2} \div \frac{5}{2} = \frac{15}{2} \times \frac{2}{5}$ $\left(\text{reciprocal of } \frac{5}{2} \text{ is } \frac{2}{5}\right)$

= 3 bottles

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